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Assessment of an integrated peat-harvesting and reclamation method: peatland-atmosphere carbon fluxes and vegetation recovery

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Abstract We document a two-year experimental trial of a recently-developed integrated peat-harvesting and reclamation technique at a poor fen in northern Ontario, Canada. We removed and conserved the uppermost ~0.3 m of peat in blocks while deeper peat was harvested from the resultant pit. We allowed the extraction pit to flood with shallow groundwater, and then reclaimed the conserved surficial peat blocks by transplanting them into the flooded pit where they formed a low, floating mat. In the 2nd year after harvest average *Sphagnum* cover in our experimental plot was intermediate (~25 %) between hummocks (~100 %) and hollows (~10 %) at an adjacent unharvested reference plot. Mean rates of *Sphagnum* productivity were greater in the experimental plot ($65\text{--}86\text{ g m}^{-2}\text{ month}^{-1}$) than in the reference plot ($45\text{--}55\text{ g m}^{-2}\text{ month}^{-1}$) for both hummock (*S.*

fuscum) and lawn (*S. magellanicum*) species, although not significantly so, indicating that the transplant had no adverse effects on *Sphagnum* health. The inundated soil conditions in the trial pit prevented the large carbon dioxide emissions that are characteristic of many harvested peatlands. During the second growing season midday net ecosystem exchange at the experimental plot was similar to that at hollows in the reference plot. However, the anoxic soil conditions in the experimental plot led to highly elevated methane emissions in both years. Our results demonstrate that the method can enable rapid re-establishment of a healthy *Sphagnum* mat and carbon dioxide sequestration function in harvested peatlands, although the global warming potential of our experimental trial was high due to elevated methane emissions.

Keywords Peat extraction · *Sphagnum* · Carbon dioxide · Methane · Peat-block reclamation · Floating peat · Acrotelm transplant

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Introduction

Dried, milled peat is widely used as a source of fuel, and as a soil additive for horticultural applications. Peat-rich nations such as Finland, Estonia, the Republic of Ireland and Russia meet sizeable proportions of their annual energy demand through the combustion of peat (Howley and Holland 2013; Joosten and Clarke

2002; Nilsson and Nilsson 2004; Turunen 2008). With approximately 1.1×10^6 km², or 12 % of its land area, covered by peatlands (Tarnocai et al. 2000; Bridgman et al. 2006), Canada is one of the world's most peat-rich nations. Approximately 24,000 ha of Canadian peatlands have been drained for horticultural use, 14,000 ha of which is currently under active extraction (Environment Canada 2014a). Interest in peat as a fuel source is also increasing in Canada given recent governmental pledges to reduce use of coal power (Tang 2009; Gleeson et al. 2006). However, peat resources and harvesting activities in Canada are almost entirely unregulated at present, leading to calls from provincial authorities for the development of regulations to reduce environmental damage (Environmental Commissioner of Ontario 2005).

Modern mechanized peat extraction techniques such as block cutting and vacuum harvesting generally involve the clear-cutting of peatland vegetation followed by the drying and harvesting of the uppermost half meter or so of peat. The removal of the live vegetation layer—particularly *Sphagnum* moss—greatly reduces the ability of peatlands harvested in this way to maintain a soil-moisture regime suitable for the re-establishment of peat-forming plants such as *Sphagnum* and *Eriophorum* (Price 1997; Rochefort 2000; Tuittila et al. 2004), leading to increased soil aeration and enhanced peat decomposition. The result is often long-term ecohydrological damage (van Seters and Price 2001), and decades or more of accumulated soil carbon can be lost in just a few years post-harvest (Waddington et al. 2002, 2009; although see Glatzel et al. 2004). Considering also diminished carbon sequestration due to the removal of live plants, harvested peatlands can become strong, persistent net sources of atmospheric carbon dioxide (CO₂) (Petrone et al. 2001; Waddington et al. 2002).

A number of studies have examined previously block-cut peatlands that have since been abandoned and allowed to revegetate spontaneously. Price and Whitehead (2001) discovered *Sphagnum* recovery was greatest in shallow ditches and absent in raised baulks. Even after several decades, recovery of *Sphagnum* mosses are limited to shallow ditch areas, with adjacent raised baulks devoid of *Sphagnum*, leaving plant communities dissimilar to those that existed pre-harvest (Elling and Knighton 1984; Yli-Petäys et al. 2007; Pouliot et al. 2011, 2012; González et al. 2013). In vacuum harvested peatlands, active restoration and

reclamation techniques such as ditch blocking (Price et al. 1998) and the moss-transplant method, where *Sphagnum* diaspores are re-introduced onto bare peat surfaces (Ferland and Rochefort 1997; Graf et al. 2012; González and Rochefort 2014), can sometimes promote the re-establishment of plant communities and may help to restore pre-harvest peat-accumulation and carbon-sequestration functions (Waddington and Warner 2001; Tuittila et al. 2004; Waddington et al. 2010). However, in many cases the ecohydrological damage caused by peat harvesting may be long lasting even in spite of restoration efforts (Waddington and McNeil 2002; Pouliot et al. 2012; González et al. 2013).

A recently-developed extraction technique called the peat-block reclamation (PBR) method (also known as wet harvesting or the acrotelm-transplant method) integrates restoration efforts directly into the harvesting method (Cagampan and Waddington 2008a, b; Waddington et al. 2009). Peat-block reclamation involves the uppermost 0.3–0.5 m or so of peat, including the live vegetation layer, being removed in blocks using a backhoe and set aside. A layer of peat between 1 and 3 m thick is then harvested from the resulting pit, before the conserved surficial peat blocks are reclaimed by transplanting them back into the extraction pit soon after harvest. By reclaiming surficial peat in this way the site may be more rapidly rehabilitated into an ecosystem capable of supporting healthy peatland vegetation, which may in turn help it to re-establish a net carbon sink function (Cagampan and Waddington 2008a) and lessen further ecosystem and soil degradation. The harvest will almost always extract peat below the water table, leaving a 1–3 m deep ditch or pool. In some cases this pool may be drained or pumped to aid the drying of harvested peat, and in other cases it may be left inundated with shallow groundwater. The conditions in the extraction pit into which the reclaimed blocks are transplanted seem likely to be important to the site's post-reclamation carbon budget, although there has yet to be an assessment of the PBR method for conditions in which the transplanted peat blocks are floated in a flooded extraction pit.

Our aim was to assess the short-term ecosystem response of an Ontario peatland, in terms of peatland-atmosphere carbon gas exchanges and changes in plant community composition, to the floating PBR technique. More specifically:

- (1) We suspected that floating the reclaimed peat blocks into a flooded extraction pit would provide a stable soil hydrological regime in the reclaimed peat, reducing water stress experienced by the transplanted vegetation mat. We wished to investigate whether this enabled a healthy plant community to be transplanted en bloc without irreparable damage, particularly the important *Sphagnum* layer.
- (2) Elevated CO₂ emissions immediately after peat harvest are a common problem in many existing industrial harvest techniques. We wished to investigate whether a flooded transplant pit would suppress CO₂ emissions through the prevalence of anoxic soil conditions in the reclaimed peat blocks.
- (3) We suspected that any reductions in CO₂ emissions may be accompanied by elevated methane (CH₄) release from a PBR transplant pit, again due to the saturated, anoxic soil conditions in the floating transplanted peat blocks. We wished to use both CO₂ and CH₄ fluxes to examine global warming potential from a PBR-harvested site.

Additionally, we measured a number of simple hydrological and physical metrics that may act as drivers of any post-harvest changes in plant community composition and carbon gas exchanges.

Methodology

Study site

The study site is a poor fen in north-western Ontario (48.83°N, 89.98°W), 52 km south-east of the town of Upsala. The site is a low-boreal wetland, located at the transition between the continental and humid boreal forest, and was selected on the basis of it being broadly representative of the kinds of peatlands that may be harvested for fuel in Ontario in future (Gleeson et al. 2006). Long-term average precipitation is 679 mm yr⁻¹ and long-term average annual temperature is 0.7 °C (Environment Canada 2014b; data for Upsala, Ontario meteorological station 1948–2009). The site is characterized by a variable topography of hummock and hollow microhabitats that is typical of many northern peatlands (e.g., Rydin and Jeglum

2006) with a peat depth of 3.9 m. Average pH of soil water immediately prior to our experiments was 5.63, while average electrical conductivity was 20.7 µS cm⁻¹. The site is dominated by mosses, shrubs and sedges, with mosses consisting primarily of *Sphagnum fuscum* (Schimp.) H. Klinggr. in hummocks and *Sphagnum magellanicum* Brid. in both hummocks and hollows. *Sphagnum angustifolium* (Warnst.) C.E.O. Jensen and *Polytrichum strictum* Menzies ex Brid. are also present. Shrubs at the site consist of *Chamaedaphne calyculata* (L.) Moench, *Andromeda glaucophylla* Link and *Vaccinium oxycoccus* (Hill) W.D.J. Koch. Sedges are limited mainly to *Carex limosa* L., while tree species (approximate density of 500 stems ha⁻¹) include *Picea mariana* (Mill.) Britton, Sterns & Poggenb. and *Larix laricina* (Du Roi) K. Koch.

Experimental design

We took our measurements in two plots: a natural, unharvested (“reference”) plot and a nearby manipulated (“experimental”) plot approximately 80 m to the east where we had performed a small-scale experimental trial of the floating PBR method immediately prior to the study period. The reference and experimental plots each measured 12 × 12 m in plan (maximum size approved following environmental impact assessment). In April 2008, a week or so before we began monitoring, we used a backhoe to extract peat from the experimental plot to a depth of 1.5 m, leaving a pit that measured 12 × 12 m in plan and which began to fill with shallow groundwater seepage from its saturated walls and floor. The uppermost layer of peat (approximately 0.3 m thick), including the live vegetation layer, was set to one side and left out on the peatland surface for a week, as it might be during an industrial-scale application of the method. The conserved blocks of surficial peat and vegetation will undoubtedly have undergone some degradation during this storage period, but we believe this to be a representative simulation of how larger-scale extractions might implement PBR. At the beginning of May 2008 we replaced the upper ~0.3 m of peat into the now flooded extraction pit according to the PBR method as described by Cagampan and Waddington (2008a, b). We constructed wooden boardwalks in both plots to allow us to take measurements with a minimum of damage to the peat and vegetation. We

measured peatland-atmosphere carbon gas fluxes, hydrological parameters and vegetation characteristics over two growing seasons, from May to August in 2008 and 2009.

In the reference plot we established six permanent sampling points for gas-flux and water-table measurements, with three each on hummocks and hollows. The peat extraction and transplant in the experimental plot caused unavoidable damage to the transplanted peat blocks and vegetation there, meaning that the pre-existing surface topography and plant communities could no longer be identified reliably. Given also the anticipated changes to water-table regime (a defining characteristic of hummocks and hollows—Belyea and Clymo 2001) in the transplanted peat, we deemed it inappropriate to attempt to identify hummock and hollow habitats in the experimental plot. We chose six permanent sampling locations in the experimental plot.

Meteorological and hydrological measurements

We installed an automated weather station (AWS) immediately to the south of the experimental plot to measure precipitation, photosynthetically-active radiation (PAR) and air temperature at half-hourly averaging periods. Technical problems with the rain gauge during the 2008 study period led to some gaps in the precipitation dataset, which we filled using data from the nearby Upsala weather station (49.03°N, 90.47°W). In each plot we installed PVC monitoring wells (inner diameter 0.05 m) next to each of the six gas-flux collars (see “Carbon gas exchange” section) in order to record water-table depths. Additionally, at the reference plot we installed a further five wells around the edge of the plot; at the experimental plot we installed a further 15 wells around the edge of the plot. Throughout the two study seasons we measured water-table depths manually at all wells at a frequency of between one and five times per week. In the experimental plot we measured water-table depth relative to the surface of the floating peat blocks, not to the base of the extraction pit, meaning that measurements were consistent between the two plots.

Carbon gas exchange

We used a closed-chamber method to measure carbon gas fluxes between the peatland and the atmosphere

(e.g., Nykänen et al. 1995; Tuittila et al. 1999). We installed square (0.6 × 0.6 m) steel collars into the peat surface at each of the six permanent sampling locations in both the reference (three hollows, three hummocks) and experimental plots. The collars extended into the peat to depths of between 0.10 and 0.15 m; the upper edges of the collars sat flush with the peat surface and consisted of a shallow tray into which we fitted our portable chambers. In order to form gas-tight seals between the collars and the chambers before each measurement we filled the collars with a shallow depth of water and then carefully fitted the chambers in place.

We measured net rates of CO₂ exchange between the peatland and the atmosphere (net ecosystem exchange, NEE) twice per week at each collar during the sampling periods using a portable, transparent, square-section acrylic chamber (footprint area 0.36 m²; height 0.3 m; internal volume 0.108 m³, see Nykänen et al. 1995). We monitored changes in concentrations of CO₂ in the chambers at 30-second intervals during three-minute sampling periods using portable infrared gas analysers (PP-Systems EGM-1 and EGM-2). We estimated CO₂ exchange rates by fitting a linear regression between time and the CO₂ concentration, where the slope of the regression gives the CO₂ exchange rate. We repeated our measurements each time using the transparent, uncovered chamber, and then with a fully opaque shroud covering the chamber to determine ecosystem respiration, ER, in the absence of PAR. We calculated gross ecosystem production, GEP, for each sampling period at each collar as the difference between the measured NEE and ER values for the same period. We opened and aired the chamber for a minimum of 30 s between measurements at the same collar to allow CO₂ concentrations in the chamber to return to background levels. We used a small internal electric fan to homogenize air humidity, temperature and gas concentrations in the chamber during sampling. A commonly used method to estimate seasonal CO₂ exchange is to measure CO₂ exchange at different PAR levels and use light response curves to model CO₂ exchange over the whole season (e.g., Tuittila et al. 2004). However, our aim was not to calculate a complete carbon budget but rather to make general comparisons between treatments. Hence we measured all collars within a few hours of one another around the middle of the day under similar light

conditions. Light conditions varied among days but were very consistent between samples taken on the same day. We took most of our samples under high irradiance ($>1,000 \mu\text{mol s}^{-1}$), conditions under which the majority of peatland plants are saturated with respect to light (Small 1972; Rice et al. 2008).

We estimated peatland-atmosphere methane (CH_4) fluxes once per week from the same collars used for CO_2 fluxes (e.g., Nykänen et al. 1995). We sealed opaque, square-section, stainless steel chambers to the collars. The steel chambers had the same dimensions as the acrylic ones used for CO_2 measurements. A small circuit of PVC tubing running out of and back into one side of the chamber acted as a septum from which we extracted gas samples using a 10 ml syringe. Once a seal was established between the collar and the chamber we opened the chambers and pumped the PVC tubing to ensure that internal CH_4 concentrations returned to background atmospheric levels between tests. We extracted gas samples from the chamber at the beginning of the test (background atmospheric CH_4 content) and after 10, 20 and 30 min in order to estimate rates of CH_4 efflux. We again used small electric fans to homogenize air temperature, humidity and CH_4 content inside the chambers and the tubing during the sampling periods. Samples were analysed for CH_4 content in laboratories at McMaster University in Hamilton, Ontario within 48 h of collection (Varian 3800 gas chromatograph with a flame ionisation detector and Porapak N column). Methane flux was calculated using the approach summarized in (Tuittila et al. 2000). We took our CH_4 samples during the middle of the day so as to allow comparison with our estimated CO_2 fluxes.

Vegetation dynamics

Vegetation cover

We surveyed vegetation cover five times between 2nd June and 10th August 2008 and three times between 27th May and 20th July 2009. We determined the proportion of the ground surface occupied by each species present inside each gas collar (see “Carbon gas exchange” section) visually at both the experimental and reference plots (see Strack and Waddington 2007). In 2009 we extended the vegetation survey with 12 quadrats ($0.3 \times 0.3 \text{ m}$) in the references plot. Six

hollows and six hummocks were selected at random distances along a 12 m transect. Because plant functional responses were in focus we did not always identify to the species level. In 2008 trees, shrubs and herbs were identified to the species level while the remaining species were grouped into grasses, sedges and mosses. In 2009 we identified mosses to the species level.

Sphagnum growth rates

We used Clymo's (1970) cranked-wire method to estimate *Sphagnum* growth rates. Cranked wires were constructed from 0.3 m long, 0.002 m diameter stainless steel rods and inserted into the moss so that one end protruded above the moss surface by approximately 0.08 m to begin with. We ensured that the cranks in each wire were buried approximately 0.02 m below the surface to prevent the wire from moving vertically. While some studies have demonstrated that crankwires may be unsuitable in areas with peat subsidence (Waddington et al. 2003a) the peat blocks at the experimental plot did not subside but rather merely floated on the water surface so we are confident as to the reliability of this method. We estimated rates of *Sphagnum* growth by measuring the decreasing distance between the moss surface and the protruding end of the wire, anchored in place by the crank (see also McNeil and Waddington 2003). We chose five *Sphagnum fuscum* hummocks and five *Sphagnum magellanicum* lawns at each plot (reference, experimental). We installed five cranked wires at each of these 20 sampling locations (25 cranked wires per species per plot; 100 wires in total) and measured monthly growth increments during a part of the second growing season (2009) only, between 27th May and 29th July. After the final growth increment measurement we took small destructive samples (0.04 m deep, $0.06 \times 0.06 \text{ m}$ across) from the surface of each cranked-wire location, including the live capitulum, to determine dry bulk density of the new moss. We then oven-dried the samples at 70°C for increments of 24 h until further drying led to no observable changes in weight. We used these final dry weights to calculate dry bulk density, ρ_B , and to convert *Sphagnum* growth increments, Δh (dimensions of length) to productivity rates, P (gravimetric rate of moss growth per unit surface area of moss) according to the method

described by Clymo (1970). We employed the following equation:

$$P = \frac{\Delta h \rho_B}{\Delta t}, \quad (1)$$

where P is *Sphagnum* productivity ($\text{g m}^{-2} \text{ month}^{-1}$), Δh is the vertical growth increment of the *Sphagnum* mat as measured against the cranked wires (m), ρ_B is *Sphagnum* dry bulk density (g m^{-3}), and Δt is the length of time between the initial and final height measurements (months, assuming a month is equal to one twelfth of a year, or 30.42 days).

Analysis

To examine the overall differences in carbon fluxes (CO_2 and CH_4) between the experimental and the reference plots (hollows and hummocks) we compared the means over the measuring period (i.e., averaging over all measurements). This comparison estimates the difference in carbon flux for the investigated sites over the dates we performed the measurements. Statistical tests of the two factors year (2008, 2009) and habitat (reference hummocks, reference hollows, experimental plot) and their interactions were performed by fitting generalized least-square models using the *nlme* package (Pinheiro et al. 2014) where we allowed for heterogeneity in variances among groups and between-year dependence within plots.

To examine if the plant community at the experimental plot followed a succession towards or away from the reference site we performed ordination analysis (a multivariate analysis where the proportional coverage of each plant species is used as a response variable). Vegetation changes over time at the reference and experimental sites were illustrated by plotting results from a Nonmetric Multidimensional Scaling (NMDS) analysis performed in *vegan* ver 2.0–7 (Oksanen et al. 2013). Uncertainty is illustrated by 95 % confidence ellipses around the treatment centroid. In addition we performed separate tests for two important groups in 2009, namely *Sphagnum* and shrubs. Here repeated measurement analyses were employed using the *nlme* package and allowing for heterogeneity in variances among habitat (reference hummocks, reference hollows, and experimental plot). We analysed *Sphagnum* production using two-way ANOVAs with species (*S. magellanicum*, *S. fuscum*) and plot (reference, experimental) as factors.

Results

Climate and hydrology

Rainfall was generally high during the study period. May 2009 experienced greater precipitation than any other May in the baseline period (100th percentile), while July 2008 was in the 98th percentile of the baseline record for July precipitation. June 2008 (91st percentile), May 2008 (81st percentile) and August 2009 (92nd percentile) also saw heavy rainfall. Only August 2008 (16th percentile) and June 2009 (11th percentile) experienced lower-than-average rainfall. Although temperature was more moderate than rainfall during the study period, it was generally cool, particularly in 2009 when only June was above average temperature (60th percentile); July 2009 was in the 7th percentile of July temperatures.

Average water-table depth was approximately 15 cm above the peat surface at the experimental plot, 6 cm in reference hollows, and 23 cm below the surface in reference hummocks. The transplanted surface in the experimental plot was highly uneven, with some areas of transplanted peat protruding above the water table, and other areas submerged. The deeply-ponded average water tables for the experimental plot shown in Fig. 1 reflect the influence of some deep pools that formed in depressions in the transplanted surface, including some gaps in between transplanted peat blocks. Magnitudes and directions of change in water table were similar between all three habitats for most of 2008, apart from the end of the season when experimental water tables deepened noticeably. Throughout the entire 2009 season average water-table depth at the experimental plot exhibited steadier behavior, with noticeably less drawdown during dry periods and less pronounced recovery after rainfall than at the reference plot (Fig. 1).

Carbon gas exchanges

Seasonal averages of midday net ecosystem exchange, NEE, at the reference plot were similar between 2008 and 2009; both hummocks and hollows showed negative values indicating a net carbon sink during the middle of the day. The experimental plot exhibited a different pattern in NEE from the reference plot (habitat: $F_{(2, 18)} = 4.9$, $p = 0.02$); although this difference between the experimental and reference

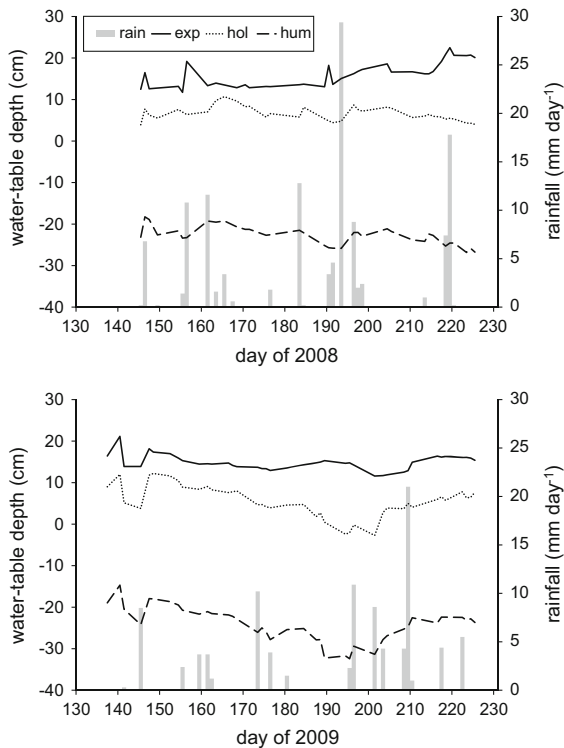


Fig. 1 Time series of water-table position (averaged within each of the three habitats; positive values represent water ponded above peat surface) during 2008 (*upper panel*) and 2009 (*lower panel*) study periods

plots decreased from 2008 to 2009, as evidenced by a significant interactive effect (year $\times$ habitat: $F_{(2, 18)} = 6.1$, $p < 0.01$). The experimental plot was a large source of CO_2 during the middle of the day in 2008 but NEE was close to zero in 2009, and therefore similar to reference hollows that year (Fig. 2a).

Ecosystem respiration, ER, declined significantly for all three habitat types from 2008 to 2009 (year: $F_{(2, 18)} = 676.6$, $p < 0.0001$). In both years ER at the experimental plot was similar to that at reference hummocks; while ER at reference hollows was significantly lower in both years (habitat: $F_{(2, 18)} = 31.3$, $p < 0.0001$). There was also a significant interactive effect in ER between factors (year $\times$ habitat: $F_{(2, 18)} = 6.3$, $p = 0.008$). However, this interaction is of less ecological interest because it appears to reflect the fact that reference hollows exhibited a smaller absolute decrease in ER between years than did the other habitat types, but a similar proportional decrease (Fig. 2b).

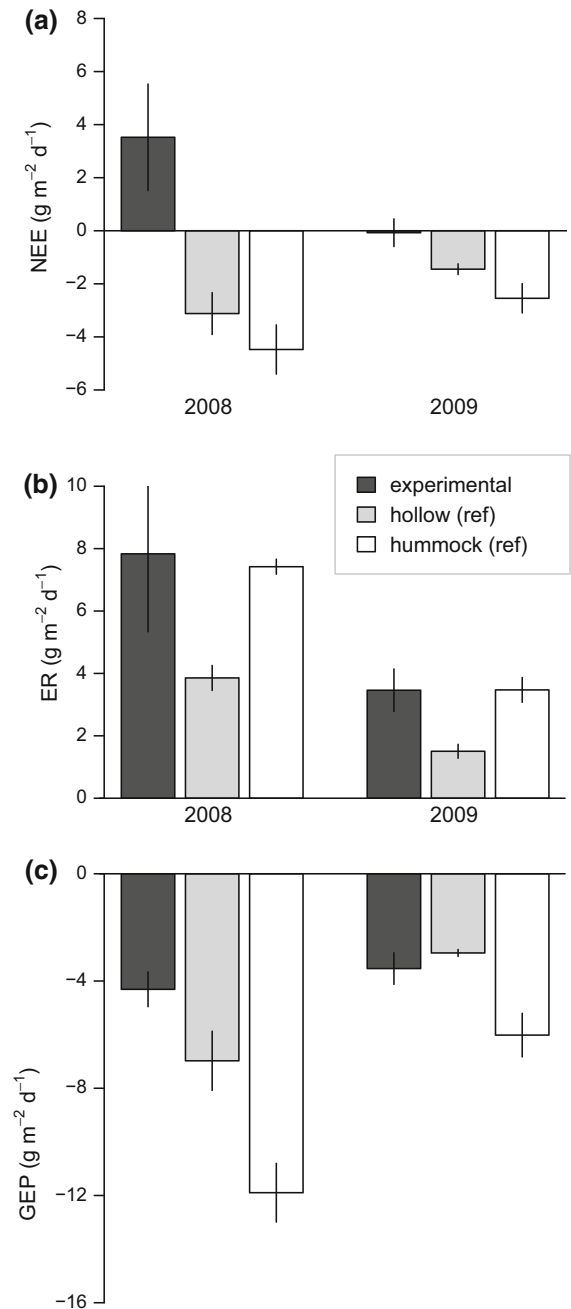


Fig. 2 Average daytime CO_2 flux rates between May 27th to August 13th (August 12th 2008): **a** net ecosystem exchange (NEE); **b** ecosystem respiration (ER); and **c** gross ecosystem productivity (GEP) at the three habitat types. Measurements were taken twice a week during the middle of the day. Error bars show standard error of the mean ($n = 3$ for reference hollows/hummocks; $n = 6$ for the experimental plot)

At the reference plot the decrease in ER between 2008 and 2009 was accompanied by a decrease in gross ecosystem productivity, GEP, for both hummocks and hollows (year: $F_{(2, 18)} = 75.5, p < 0.0001$). However, the experimental plot did not follow this pattern and exhibited similar GEP in both years (habitat: $F_{(2, 18)} = 12.92, p < 0.0001$; Year $\times$ Habitat: $F_{(2, 18)} = 23.3, p < 0.001$). In 2009 GEP at the experimental plot was similar to that seen in reference hollows (Fig. 2c).

Methane emissions at the reference plot were similar between the two habitat types there (hummock, hollow) and between years. Methane emissions from the experimental site were much greater (habitat: $F_{(2, 18)} = 18.8, p < 0.0001$), and unlike the reference plot increased substantially from 2008 to 2009 (year $\times$ habitat: $F_{(2, 18)} = 23.3, p < 0.001$) so that in 2009 the measured fluxes were more than 10 times greater than at the reference plot (Fig. 3).

We used our measured NEE and CH_4 fluxes to estimate a global warming potential (GWP) for the reference and experimental plots during daytime conditions. For the reference plot we averaged the

fluxes from the two habitat types there (hummock, hollow) to give values that are representative of the plot as a whole. For both reference and experimental plots we assumed that a unit flux of CH_4 has a GWP equal to 28 times that of CO_2 , representative of the behavior of these greenhouse gases in the atmosphere over a timescale of 100 years (ignoring climate-carbon feedbacks: Myhre et al. 2013). Daytime fluxes at the reference plot had a small net cooling effect in both years, as evidenced by negative midday GWP values; and a strong warming effect in both years at the experimental plot, which largely reflects its high CH_4 emissions (Table 1).

Vegetation dynamics

Average *Sphagnum* cover at the experimental plot collars varied between 22 and 35 % in the summer of 2009, which was higher than reference hollows (8–19 % cover) but much lower than reference hummocks (close to 100 % cover). A similar trend was observed for the shrub cover: with between 5 and 10 % shrub cover at reference hollows, 15–20 % at the experimental plot, and up to 50 % at reference hummocks by the end of the summer (habitat effect: $F_{(2, 21)} = 6.3, p = 0.007$).

Our NMDS analysis for 2008 shows that plant community composition at the experimental plot was initially similar to reference hummocks and followed a similar trajectory over the first summer (Fig. 4a). However, mosses were aggregated in the 2008 survey and there might have been a larger difference in the NMDS if *Sphagnum* species had been separated from true mosses since the experimental site had a substantial amount of *Polytrichum* in 2008. At the start of the 2009 study season the experimental site was still close to reference hummocks, but then began to shift towards reference hollow communities

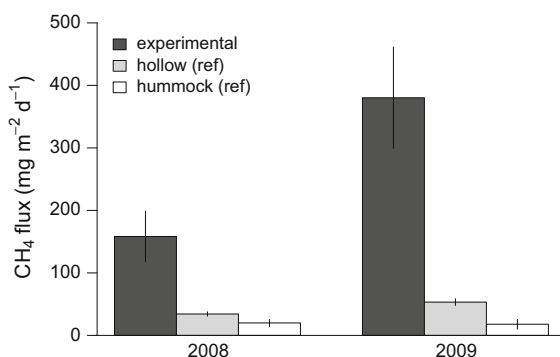


Fig. 3 Seasonal averages of daytime methane emissions from the three habitat types. Error bars show standard error of the mean

Table 1 Summary of global warming potential, GWP, and its components during the middle of the day for average measured fluxes at the experimental and reference plots

Plot	Average midday NEE (g CO ₂ m ⁻² d ⁻¹)		Average midday CH ₄ flux (g CO ₂ -e m ⁻² d ⁻¹)		Average midday GWP (g CO ₂ -e m ⁻² d ⁻¹)	
	2008	2009	2008	2009	2008	2009
Experimental	3.50	−0.07	4.43	10.64	7.93	10.57
Reference	−3.80	−2.00	0.76	1.00	−3.04	−1.00

Positive values indicate a net warming effect; negative values indicate a net cooling effect. Values for methane fluxes and daytime GWP are given in units of CO₂-equivalence, assuming methane has 28 times the global warming potential of CO₂ (Myhre et al. 2013)

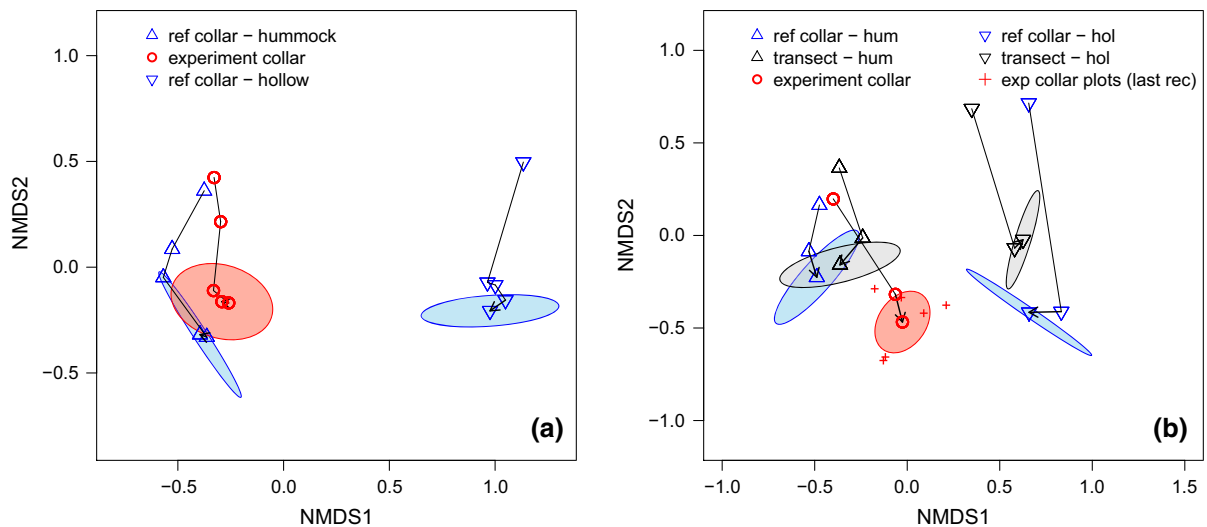


Fig. 4 Vegetation changes illustrated by an ordination (non-metric multidimensional scaling, NMDS) at the reference (ref) and experimental (experiment) plots in: **a** 2008, and **b** 2009. Points are centroids (means of the first two axes) for each group. Points close to each other have similar plant communities. Lines

show the movement between survey dates over the summer and filled ellipses are 95 % confidence intervals at the final survey in August. In 2009 additional quadrants were surveyed along a transect. The individual collar points at the experiment site are shown for the final survey in 2009

(Fig. 4b). By the end of the study period the plant community at the experimental plot was in between reference hummocks and hollows. The added transect quadrats were very similar to the collars indicating that the location of the collars were representative for the reference plot.

Calculated mean rates of *Sphagnum* production were greater at the experimental plot for both *S. fuscum* ($85.9 \pm 15.9 \text{ g m}^{-2} \text{ month}^{-1}$; error term represents standard error of the mean) and *S. magellanicum* ($64.7 \pm 10.9 \text{ g m}^{-2} \text{ month}^{-1}$) than for either species at the reference plot (*S. fuscum*: $45.4 \pm 11.8 \text{ g m}^{-2} \text{ month}^{-1}$; *S. magellanicum*: $54.5 \pm 16.9 \text{ g m}^{-2} \text{ month}^{-1}$). However, this difference between plots was not statistically significant ($F_{(1, 16)} = 4.1$, $p = 0.09$); neither a species effect nor an interaction effect were detected (species: $F_{(1, 16)} = 0.21$, $p = 0.66$; plot $\times$ species: $F_{(1, 16)} = 1.2$, $p = 0.30$).

Discussion

Vegetation dynamics

Our results indicate that *Sphagnum* was growing well during the 2009 study season, and that it was not declining in coverage during the season. Average

Sphagnum productivity was greater at the experimental plot than at the reference plot; even though these differences in mean growth rates were not significant this is a highly encouraging result because it indicates that the experimental treatment had no detrimental effects on the growth and recovery of the surviving *Sphagnum* in the 2 years since harvest. The health of the surviving *Sphagnum* seems to have been facilitated by the wet conditions there, as well as perhaps the higher than average rainfall in both years. Flooding of basins or trenches in previously block-cut peatlands has been seen to invigorate *Sphagnum* growth on timescales similar to our study, including dry-niche, hummock-forming species such as *S. fuscum* (Roche-fort et al. 2002) and also over decadal timescales (Bönsel and Sonneck 2011; González et al. 2013). The moss-transplant method often uses straw mulch to protect *Sphagnum* diaspores during early stages of regrowth by shading and improving near-surface moisture retention (Price et al. 1998). Moss transplants sometimes also use phosphorous fertilizer to encourage recolonization by *Polytrichum strictum* (Roche-fort et al. 2003; González and Rochefort 2014). Our results illustrate an example of how a healthy moss mat can be encouraged to re-establish rapidly after peat harvesting without the need for straw mulch or fertilizer.

The peat and plant communities that we studied at the experimental plot comprised a mixture of what had been hummock and hollow habitats prior to peat extraction. Because the experimental plot contained a substantial amount of *Sphagnum* our sampling locations there resembled hummocks early in the season; after the regrowth of vascular plants (mainly graminoids) later in the season plant community composition at the experimental plot was in between reference hummocks and reference hollows. Again, this may also reflect in part the high rainfall received, particularly in the second half of the 2009 growing season. Some collars in the experimental plot became more like hummocks and others more like hollows, providing a possible pathway towards re-establishing a characteristic hummock-hollow microtopography and biodiversity (Rydin and Jeglum 2006). The re-establishment of hummock-hollow microtopography pattern can take more than a century in abandoned cut-over peatlands, but there is evidence that this recovery time might be reduced to between 10 and 30 years if restored using the moss-transplant (diaspore dispersal) method (Pouliot et al. 2011). Our results suggest that the PBR method could further enhance the recovery of microtopography. This is likely possible due to the patchiness of *Sphagnum* in combination with partial cover of woody shrubs that is essential for hummock formation (Pouliot et al. 2012), as well as abundant patches of open water in between providing suitable habitats for colonization by hollow species (Landry et al. 2012).

Greenhouse gas fluxes

The low ecosystem respiration at our experimental plot, and the near-zero net ecosystem exchange during the second study season, are causes for optimism for implementation of the PBR method in flooded extraction pits. Carbon dioxide emissions are often several times higher from harvested sites than from natural sites (Waddington and Price 2000; Waddington et al. 2002). Previous studies have observed that CO₂ emissions from harvested peatlands are suppressed under anaerobic soil conditions such as in flooded former block-cuttings, where shallow water tables and anoxic soil conditions inhibit aerobic respiration (Waddington et al. 2001; although see Basiliko et al. 2007). This appears to have been the case at our experimental plot. Additionally, the

exposure at the surface of older, more decomposed peat presents decomposers with a less labile substrate. This phenomenon has been cited in the past for reduced respiration and CO₂ emissions in harvested peatlands (Glatzel et al. 2004; see also Belyea 1996), and may also have played a role at our experimental site. Glatzel et al. (2004) argued that the same quality effect could explain reductions in methane emissions in incubation experiments where deeper peat layers had been exposed, but only if water tables were deep. However, the inundation of our experimental plot with shallow groundwater appears to have more than counteracted any such peat-quality effect and led to high CH₄ emissions.

As expected, the high CH₄ emissions from our experimental plot represent a major drawback to the PBR method in a flooded extraction pit. Methane fluxes dominate the midday carbon budget of our experimental plot by accounting for the majority of its large midday GWP values (Table 1), particularly during the 2009 study season. Where block-cut peatlands are not actively restored, methane emissions can be an order of magnitude lower in harvested bogs than in natural sites, even if CO₂ fluxes are greatly enhanced (Sundh et al. 2000; Waddington and Price 2000; Maljanen et al. 2010), and in some cases drainage can even convert peatlands from sources to sinks of methane (Lohila et al. 2011; Cooper et al. 2014). Our trial of the PBR method in a flooded pit therefore compares unfavorably in this regard to unrestored peatlands. However, attempts to restore peatlands harvested using traditional means often involve the flooding of block-cut trenches and drainage ditches, for example, leading to anoxic soil conditions and high methane fluxes that are comparable to our measurements (Tuittila et al. 2000; Waddington and Day 2007; Wilson et al. 2009). Also, unlike moss-transplant methods, PBR does not use straw mulch to protect *Sphagnum* diaspores during early regrowth. The decomposition of mulch can add considerably to the greenhouse gas fluxes from moss-transplant sites (Waddington et al. 2003b) and are accounted for in carbon life-cycle analyses of peat harvesting methods (Waddington et al. 2009). The greater-than doubling of average CH₄ emissions at our experimental plot between 2008 and 2009 is consistent with research by Strack and Zuback (2013) who reported prolonged elevation of methane

fluxes 10 years after harvest at a Canadian treed bog. Our second study season was not warmer than the first, so the well-documented relationship between peat temperature and methanogenesis (e.g., Moore and Dalva 1993; Dunfield et al. 1993; although see Kang and Freeman 2002) offers no explanation for the inter-annual increase. We suspect that some of the increased CH₄ emissions may have been caused by anaerobic decomposition of recently fixed, high quality carbon from vigorous vegetation regrowth, possibly including plant root exudates (cf. Glatzel et al. 2003; Marinier et al. 2004).

Hydrology

Most of our main findings—namely the apparent short-term success of *Sphagnum*, low net ecosystem exchange of CO₂, and high methane emissions—can be explained by differences in the hydrological regimes between the experimental and reference plots. The stability of the water table at our experimental plot for much of the study period is likely a direct result of transplanting the surficial blocks into a flooded extraction pit. González and Rochefort (2014) showed how a hot, dry summer immediately after restoration efforts using the moss-transplant method (scattered moss diaspores covered by straw mulch) can hinder the regrowth of healthy *Sphagnum*. Both of our study seasons were relatively cool and wet, so direct comparison is difficult. However, the availability of open water and the stability of the water table at the experimental plot seem likely to provide a buffer against water-table drawdowns during dry summers (e.g., Price 1997), and may offer an advantage over the moss-transplant method.

Drainage and subsequent rapid oxic decomposition have been shown to lead to changes in peat hydrophysical properties in unrestored block-cut or vacuum-harvested sites (Price 2003). The closure of pore spaces results in a loss of water storage capacity and so a greater range of diurnal and seasonal water-table fluctuations than in natural sites. Consequently, water tables can fall into deep peat layers due to lack of storage capacity (Price 1997; Schouwenaars 1993), which may in turn inhibit successful re-establishment of *Sphagnum*. Cagampan and Waddington (2008b) demonstrated

that transplanting surficial peat blocks into a PBR extraction pit leads to a more stable soil moisture regime, which seems to have facilitated the success of *Sphagnum* at our experimental plot and is likely to avoid problems associated with changes in peat hydrophysical properties.

Conclusions

Our study illustrates some potentially important advantages of implementing the peat-block reclamation method in a flooded extraction pit compared to other peat harvesting and restoration techniques. By the second growing season our trial of the method was close to CO₂-neutral during the middle of the day and its midday net ecosystem exchange was statistically indistinguishable from the adjacent unharvested plot. Ecosystem respiration was similar to that from undisturbed hummock habitats indicating that the method had been successful in suppressing the high CO₂ emissions common in unrestored sites. Our results also indicate that transplanting surficial peat blocks back into a flooded extraction pit may be a viable means of facilitating the rapid re-establishment of a healthy plant community after harvest. In the second season after the transplant both wet- and dry-favoring *Sphagnum* species were growing vigorously in the flooded conditions, and midday values of gross ecosystem production were similar to those under natural conditions.

However, our trial implementation also illustrated an important drawback of the PBR method—highly elevated methane fluxes compared to natural conditions. In assessing the viability of PBR in flooded pits as an environmentally sound method of large-scale peat extraction, its potential for such high methane emissions must be carefully weighed against any potential advantages compared to other restoration methods.

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